

Dynamically Reconfigurable Paper

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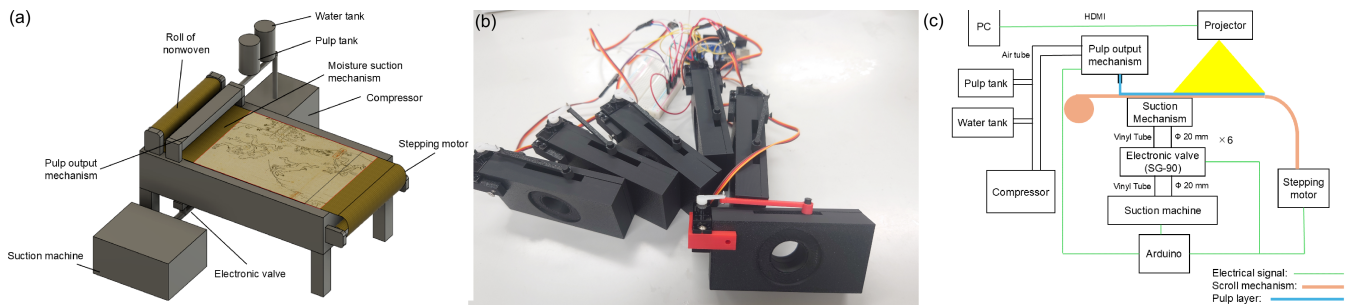


Figure 1: (a) The proposed system, (b) Electronic valve (c) Overall schematic diagram

Abstract

Paper has long been an indispensable part of our lives, and various types of paper are used in both everyday situations and in specialized fields, regardless of the field. However, paper needs to be used for different purposes, and we believe that this limitation is due to the fixed physical parameters of paper. Therefore, we focused on the fact that the traditional manufacturing process of paper has a unified process of dissolving pulp in water and drying it, regardless of the physical properties of the paper. By combining the papermaking process with an extension technique that regulates the moisture content of the paper, we were able to vary the physical parameters of the paper so that our paper is dynamic and possesses multifunctional properties not found in conventional paper. In this paper, we propose an application of our new paper's functionality to improve the interaction between computers and humans.

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1 Introduction

Paper is an indispensable part of our lives and is used for various types of purposes. On the other hand, digital devices such as cell phones and personal computers, which have been rapidly developing in recent years, have various functions in a single device. We believe that the issue lies in the fixed physical parameters of paper, such as thickness, material, and moisture content, which limit its use depending on the application. Washi is manufactured by dissolving the raw pulp in water, straining, and drying. This process is standardized regardless of the nature or thickness of the paper. Therefore, we focused on this traditional washi manufacturing process. The process of drying and dehydrating paper can be reversibly retrograded by adding water again. Furthermore, the combination of the suction surface that we have created for moisture suction enables finer control of moisture content, and we believe that the new paper we propose has a dynamic nature not found in conventional paper, which can handle thickness, properties, and hardness as variable parameters. We believe that this new paper, like modern digital devices, can perform a variety of functions with a single piece of paper, and that this feature will contribute to the improvement of human-computer interaction using paper. In this paper, we present three concrete proposals for new ways to interact with computers using this new paper, and the results of each proposal.

2 Related Research

Much research has been done in HCI to modify the physical properties of paper. Stockinger et al. [Stockinger et al. 2017] proposed a

low-cost paper-based capacitance sensor with ink-jet printed electrodes; Jiuke et al. [Mu et al. 2015] proposed a water-driven actuator using graphene monolayer paper; Yoshikawa et al. [Yoshikawa et al. 2024] proposed a water-driven actuator using graphene monolayer paper; and Kato et al. proposed a water-driven actuator using graphene monolayer paper; Yoshikawa et al. [Yoshikawa et al. 2024] used conductive patterns printed on paper to manipulate water droplets. Although the high responsiveness of paper to moisture has been applied to various fields, we believe that it is an unexplored field to focus on the fixed physical parameters of paper and make them variable by adjusting the moisture content, and to verify the application of paper, which has become dynamic and multifunctional, to improving computer-human interaction.

3 Prototype

The prototype is shown in Figure 1(a). The enclosure consists of a scroll mechanism, a suction surface, a pulp output mechanism, and a suction machine. In order to achieve dynamic paper, the scroll mechanism uses a stepper motor to move a long filter paper from the roll at the top of the enclosure to the roll at the bottom at a constant speed. Next to the upper roll is a pulp output mechanism that sprays water and pulp pressurized by a compressor onto the filter paper in a desired pattern. A suction surface is located at the bottom of the filter paper, which sucks only water from the pulp on the filter paper and fixes the paper on the filter paper. The suction surface consists of six sections, and each section is equipped with an SG-90 electronic valve Figure 1 (b) to control the surface on which suction is applied. By switching suction, it is possible to prevent the reduction of suction power and enable partial suction. A water-suction-capable suction unit is installed at the bottom of the main unit, and suction is applied through six separate electronic valves. The stepping motor, electronic valves, and pulp ejection mechanism were controlled using an Arduino control software Figure 1 (c), and the code was written in Python.

4 Application Scenarios

- (1) Interactive Storytelling. Experiential Storytelling with Scrolls 1(c) allows readers to have a more immersive experience by dynamically changing the texture, properties, and strength of the paper according to the progression and content of the story. For example, the paper becomes stiffer during tense scenes, allowing users to directly experience the tension, emotions, and atmosphere of the story through physical property changes, and we believe that this experience reinforces the interactivity between the paper and us.
- (2) Responsive Paper. This application provides a mechanism whereby the paper itself responds in real time to what the user writes on it. The paper's writing feel, texture, and even visual effects change to give the user a unique sensation. For example, when the user writes a negative word, the surface gradually becomes rough and uncomfortable to write on, and when the user writes a specific word such as love or joy, an embossed heart appears. The paper can return emotional feedback to the user. The paper provides emotional feedback to the user, and the act of writing itself leads to self-expression and the organization of emotions.

- (3) User-Reconfigurable Puzzle. This application allows users to experience the sensation of solving a digital puzzle using real paper. The user reconstructs the puzzle by actually cutting, tearing, attaching, and folding the puzzle pieces projected on the paper. This application enables users to have a new puzzle experience that fuses virtual and real space. Furthermore, by linking changes in the physical properties of the paper to the properties of the pieces, we believe it is possible to provide a game experience that requires users to understand the physical properties of the paper and use them to solve the puzzle in an intuitive and strategic manner.

5 Future Work

In this paper we proposed the use of a new dynamic paper for improving human-computer interaction. We will implement the three proposed applications and validate user evaluations, and we will look to improve the accuracy of the dynamic control technology and diversify the materials. We also hope to explore new application areas for our proposal through user evaluations.

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